Forest Buddy

John Allen[PM,2], Joshua McGee[CS,2], Landon Garrett[1], Gerardo Hernandez[2], Alten Reeves[2], Lane Williams[2]
Dr. Dylan Boyd[A,1]

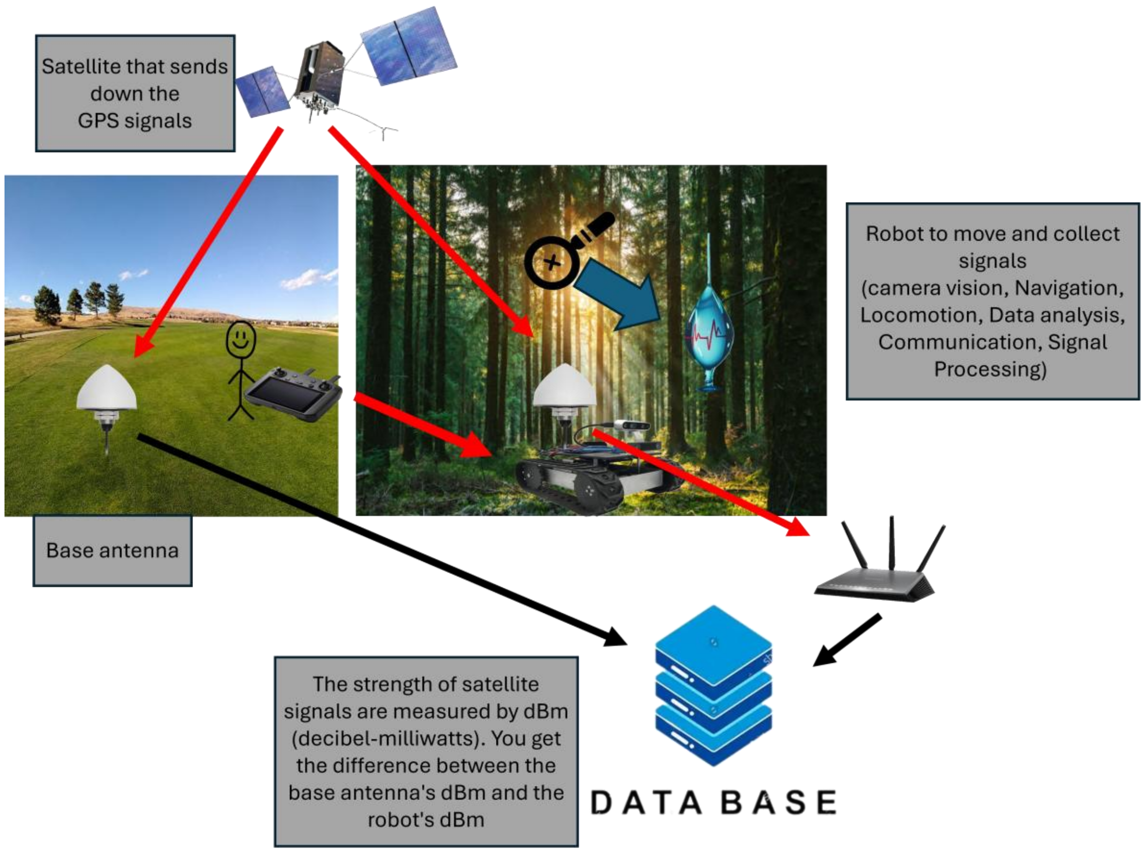
[PM] – Project Manager
[CS] – Chief Scientist
[A] – Faculty Advisor

[1] – Computer Engineering
[2] – Electrical Engineering

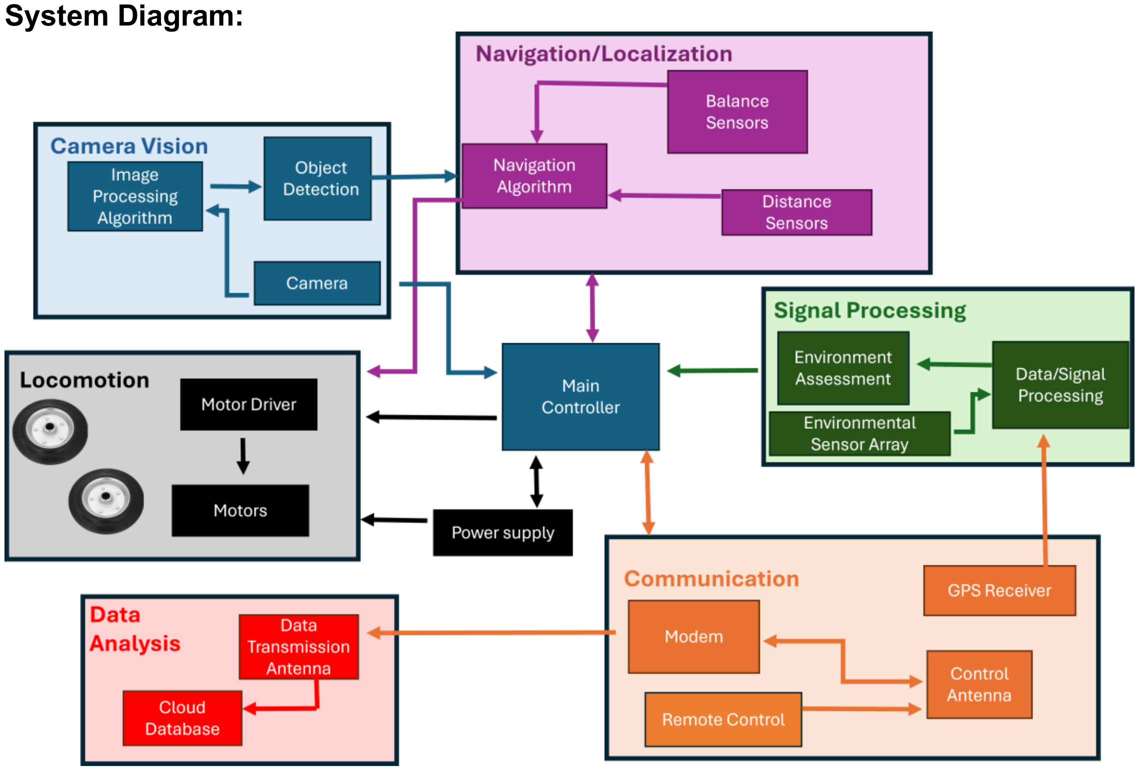
The Problem:
Nearly 11% of forested land is currently exposed to drought, with that number being expected to increase to 38% by mid-century, caused by a lack of adequate water and disease. Forests are a vital ecological pillar of many temperate climates around the globe and are especially critical in North America; they house countless ecologically significant species of both plants and animals alongside providing vital resources such as lumber and paper. Forests additionally capture nearly 20% of expelled carbon dioxide globally and provide recreation and education through America's state parks, with nearly a third of all wild spaces in the United States' mainland being forested. Preexisting methods for monitoring the health of American forests involve foresters manually assessing the health of trees by estimating the water content of the leaves or cutting down the trees to record their biomass. These methods are time consuming, resource heavy, and labor intensive. As such, there exists a need for a robust system for monitoring tree populations that can establish full scale diagnostic assessments of a forest's health, establishing both healthier forests and more efficient management of their resources.

Proposed Solution:
The Forest Buddy is a guided tele-operated robot designed to measure tree health. The proposed device will consist of several interconnected subsystems that will navigate forest terrain, gather data from the trees, and record the results. Firstly, the Computer Vision subsystem uses cameras and depth sensors to track the operator and create waypoints to produce a path the robot can take through the forest. The Navigation/Localization subsystem will consist of a multitude of sensors that will provide information on distance from objects, orientation to ground, and location which will be used to follow the created path. The Locomotion subsystem controls the robot's motors and steering to execute movement through the forested terrain. Once the robot has reached its destination, the Signal Processing subsystem will collect the GPS signals from the antenna on the robot to extract relevant signal characteristics that will be used to determine the trees' water content, it will also use cameras to record the density and health of the surrounding trees. The Data Analysis subsystem will take the data collected from the robot and compare it to the signals from the base station to estimate the water contents of the tree and use a cloud-based database to send this data back to the operator. Finally, the Communication subsystem will provide control of the robot through a remote-control if it gets stuck, alongside wireless communication between the robot and base station. Together, these subsystems will create a complete picture of the health of the environment.

Process Diagram:



System Diagram:



Team Charter

Forest Buddy

Forest Buddy
Simrall Maker Space Lab **OR** *Simrall 435*

Department / Company	**Advisors**	
ECE	**Dr. Dylan Boyd**	**boyd@ece.msstate.edu**
Expected Meeting Time & Frequency w/ Advisor(s): **Wednesday, 2 PM in Simrall 435**		

Membership:

Project Manager	
John Allen	
JPA241	EE

Chief Scientist	
Joshua McGee	
JHM502	EE

Other Team Members

Landon Garrett	LPG79	CPE
Gerardo Hernandez	GH865	EE
Alten Reeves	ATR255	EE
Lane Williams	LW1750	EE

Mode & Frequency of Communication:

We will communicate using group text messaging and regular team meetings 1-2 times per week. The team will also communicate using the Group Me app. These days will be on Tuesday at 3:30 PM and Thursday at 3:30 if needed. These meetings will take place in the Digital Media Center or Presentation Room 3060 at the Mitchell Memorial Library. The team is also meeting with Dr. Boyd on Wednesday at 2PM in Simrall 435. We will use Microsoft Teams to host common files as well as a GitHub to host code, at https://github.com/Landon-Garrett/Forest-Buddy.git.

Submission Schedule:

The team's approach to deadlines is to meet each week and discuss future assignments. We will set goals for each assignment based on when they are due. These goals include submitting assignments at least 1 day before the assignment deadline. We will have the assignment completed 2-3 days before the due date, but we will have a meeting to discuss the assignment to make sure it's completed. These goals may consist of completing weekly documents, research, or implementation of subsystems.

Team Roles:

Project Subsystems			
1	**Navigation/Localization**	Lane Williams (lead)	Alten Reeves (support)
	After a path has been mapped out, it will follow this path and detect any issue with it.		
2	**Computer Vision**	Landon Garrett (lead)	Lane Williams (support)
	Detect the operator and modify movement inputs to follow behind him.		
3	**Locomotion**	Alten Reeves (lead)	Landon Garrett (support)
	How the robot moves through the terrain and the building of the chassis		
4	**Communication**	John Allen (lead)	Gerardo Hernandez (support)
	Receives the GPS signals, relays to base station and controls the robot		
5	**Signal Processing**	Joshua McGee (lead)	John Allen (support)
	Gather and process the signal information		
6	**Data Analysis**	Gerardo Hernandez (lead)	Joshua McGee (support)
	Send the data to a cloud-based database for visualization		

Potential Obstacles:

A key consideration is that the chassis must be finished before any of the other components can be readily tested; as such, the timeframe for getting the chassis done poses a critical obstacle. Data collection must be completed before the data processing and signal processing subsystems can be worked on. Proper navigation also seems like a challenge, as navigating properly through uneven terrain is essential to not damaging any components. We anticipate some difficulty with establishing needed communication links and dealing with signal noise and multipath as consistent communication is needed for data analysis. Power management is also important, as running multiple microcontrollers and receivers, while driving motors at the same time, is power intensive, and improper management could negatively affect Forest Buddy's reliability. Logistically, we anticipate trouble with making the robot robust enough to test in appropriate environments and testing with fluctuations in weather. We anticipate some financial concerns with being able to finance a sufficiently accurate communication module and a high-gain antenna that are strong enough while being under budget. Additional financial concerns can be raised by the nature of how expensive testing could become with Forest Buddy, and how to ensure nothing breaks when testing in a forest. Finally, with all the elements of the robot being so highly integrated, further obstacles may arise from adjusting the parameters with any one subsystem.

Conflict Resolution:

Team conflicts will arise throughout the design process. If a specific team member is persistently causing conflict or falling behind, the project manager and chief scientist will first approach the member directly to understand if there are underlying issues causing the problem. If the reasons given aren't adequate (per the project manager's discretion) to explain the prolonged issue, the team will begin requiring that member to submit progress updates every day at noon to the project manager and subsystem support for their subsystem. If problems are not resolved within 5 days, the project manager will bring the issue to the faculty advisor and ask for guidance on how to proceed. If the member does not make improvements in 2 days following the faculty advisor meeting, the project manager will contact the capstone professors directly to explain the situation. If a team member misses consecutive meeting unannounced or goes 4 weekdays of not responding to required communication or not creating meaningful contributions, the project manager will escalate directly to the capstone instructor. If the team runs into technical design disagreements, each member with a stake in the decision will put together a pros-and-cons list along with their reasoning for why their approach is preferable. These lists will be presented and discussed at the next regular team meeting time, followed by a blind team vote. In the event of a tied vote, the chief scientist will act as the deciding vote. Additionally, the project manager reserves the right to call a blind vote, escalate to the faculty advisor, or escalate to capstone instructor whenever deemed necessary.

Additional Comments or Concerns:

One of the team's concerns is the similarity to a previous team's project: Leaf Link. They planned to use the robot dog to house their components but built a pulley system instead. The Forest Buddy team is going to build their own chassis to house components. The team also has concerns that, given it houses the rest of the components, the chassis must be built before any real work is done on any other subsystem. The weather also dictates when we can collect data. The rain will keep us from being able to demonstrate our project and show the data that we collected. Dr. Boyd will be funding all the pieces and products that we will use to make the Forest Buddy project.

[Signatures on the Next Page.]

Signatures

Team Signatures

John Paul Allen	09/02/2026	Joshua McGee	09/02/2026
Signature	Date	Signature	Date
John Allen		Joshua McGee	
Project Manager		**Chief Scientist**	
Landon Garrett	09/02/2026	Gerardo Hernandez	09/02/2026
Signature	Date	Signature	Date
Landon Garrett		Gerardo Hernandez	
Member		**Member**	
Alten Reeves	09/02/2026	Lane Williams	09/02/2026
Signature	Date	Signature	Date
Alten Reeves		Lane Williams	
Member		**Member**	

Advisor Signatures

	09/02/2026
Signature	Date
Dr. Dylan Boyd ECE	
Faculty Advisor	